

Engineering Mechanics

Lecture 01: Introduction

By
Dr. Aali Pant
Assistant Professor
Civil and Infrastructure Engineering
Indian Institute of Technology Jodhpur

Course Title:	Engineering Mechanics	Course Code:	MEL1010
Department:	Mechanical/CIE	L-T-P [C]:	2-1-0 [3]

English lectures

Monday: 08:00 AM – 08:50 AM

Wednesday: 08:00 AM – 08:50 AM

Non-English lectures

Monday: 05:00 PM – 05:50 PM

Wednesday: 01:00 PM – 01:50 PM

- **Statics [10 Lectures]:** Basic dimensions in Mechanics, Law of dimensional homogeneity, Vector and Scalar Quantities, Elements of vector algebra. Moment of a force about a point/axis, Couple, Free Body Diagram, Equations of Equilibrium, Static indeterminacy, Trusses and Beams, Coulomb Friction, Surface contact friction, Transmission of power through belt. Pappus-Guldinus Theorems, second moments and product of Area, Principal axes.
- **Dynamics of Particles [6 Lectures]:** Kinematics of particles, Velocity and acceleration in terms of path variables, simple relative motion, motion of particle relative to a pair of translating axes Newton's laws of rectangular coordinates/rectilinear translation, cylindrical coordinate/Central force motion. Conservation of Mechanical Energy, Work-energy equations, Center of mass based Kinetic energy, Impulse and Momentum relation of particles, Moment of momentum equations-single particle/system of particles
- **Dynamics of Rigid bodies [12 Lectures]:** Translation/Rotation of rigid bodies, Chasles' theorem, time derivative of vector for different references. Parallel axis theorems, Rotational Pure rotation of a body of revolution about its axis of revolution/combined with translation. Three dimensional rotation, moment of inertia tensor, relation between angular momentum and torque in three dimensions, Motion in non-inertial frames, centrifugal force, Coriolis force/acceleration, rate of change of vector in inertial and rotating frames.

• Reference Books

1. Shames, I., (2003), *Engineering Mechanics*, Prentice Hall
2. Gross, D., Hauger, W. and Schrder J., (2012), *Engineering Mechanics*, Springer
3. **Meriam, J. L., Kraige, L. G., (2002), *Engineering Mechanics*, John Wiley & Sons**
4. Harbola, M K., (2013), *Engineering Mechanics*, CENGAGE Learning

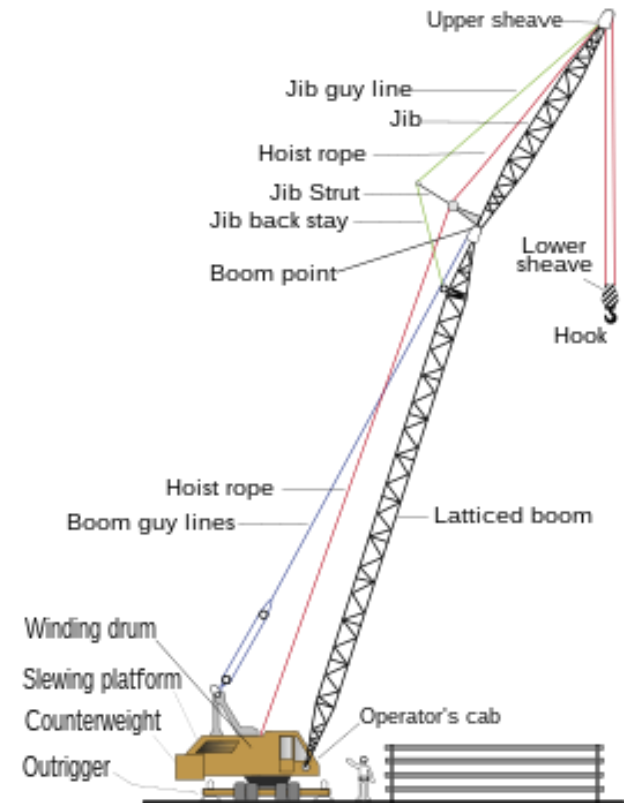
• Self-Learning Material

1. Harbola, M K., *Engineering Mechanics*, NPTEL Course Material, Department of Physics, Indian Institute of Technology Kanpur, <https://nptel.ac.in/courses/115104094/>
2. Lewin, W. H. G, *Physics I: Classical Mechanics* - Fall 1999, MIT, http://videolectures.net/mit801f99_physics_classical_mechanics/

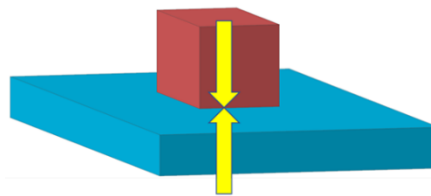
- **Mechanics** is the physical science which deals with the effects of forces on objects.
- State of rest or motion of bodies/ objects, that are subjected to the action of forces
- Sub-divided into three branches
 - a) Rigid-body
 - b) Deformable-body
 - c) Fluid-mechanics

- **Space** is. the geometric region occupied by bodies whose positions are described by linear and angular measurements relative to a coordinate system. For 3-D problems, 3 independent coordinates are needed. For 2-D problems, only 2 coordinates are required.
- **Time** is the measure of the succession of events and is a basic quantity in dynamics. Time is not directly involved in the analysis of statics problems.
- **Mass** is a measure of the inertia of a body, which is its resistance to a change of velocity. Mass can also be thought of as the quantity of matter in a body. The mass of a body affects the gravitational attraction force between it and other bodies. Mass appears in many applications in statics.
- **Force** is the action of one body on another. A force tends to move a body in the direction of its action. The action of a force is characterized by its **magnitude**, by the **direction** of its action, and by its **point of application**.

- A **particle** is a body of negligible dimensions. In the mathematical sense, a particle is a body whose dimensions are considered to be near zero so that we may analyze it as a mass concentrated at a point. We often choose a particle as a differential element of a body. We may treat a body as a particle when its dimensions are irrelevant to the description of its position or the action of forces applied to it.
- **Rigid body.** A body is considered rigid when the change in distance between any two of its points is negligible for the purpose at hand. For instance, the calculation of the tension in the cable which supports the boom of a mobile crane under load is essentially unaffected by the small internal deformations in the structural members of the boom.



- A. **Static**: Deals with equilibrium of bodies, either in rest or motion (with a constant velocity)
- B. **Dynamic**: Deals with the accelerated motion of bodies

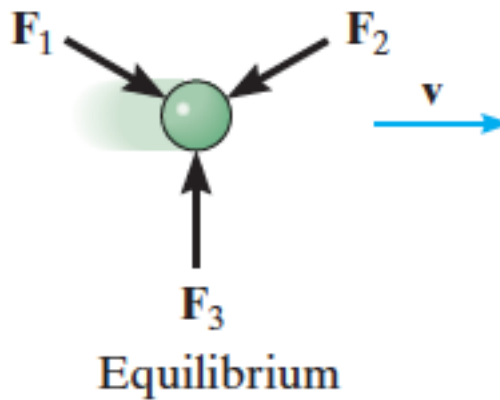


Statics



Dynamics

- A particle originally at rest, or moving in a straight line with constant velocity, tends to remain at **rest or continues to move with *uniform velocity*** if there is **no unbalanced force** acting on it.



Statics

- The **acceleration** of a particle is **proportional to the vector sum of forces** acting on it and is **in the direction of this vector sum**.

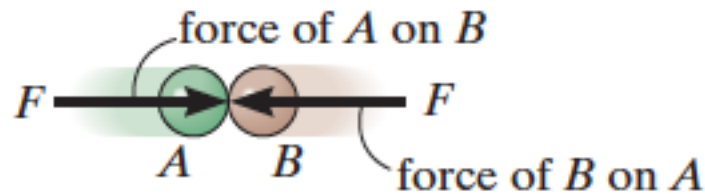
$$F = ma$$



Accelerated motion

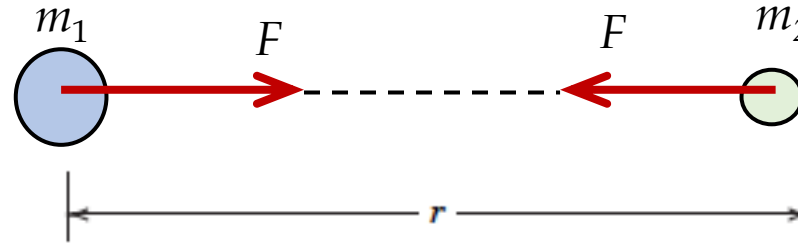
Dynamics

- The forces of action and reaction between interacting bodies are equal in magnitude, opposite in direction, and *collinear*.



Action – reaction

Constraint Forces



$$F = G \frac{m_1 m_2}{r^2}$$

Where,

F : the mutual force of attraction between two particles

G : a universal constant known as the constant of gravitation, $6.673(10^{-11}) \text{ m}^3/(\text{kg} \cdot \text{s}^2)$, based on experimental evidence

$m_1 m_2$: the masses of the two particles

r : the distance between the centres of the particles

$$W = mg, \quad \text{where } g = \left(G \frac{M}{r^2} \right)$$

Where

G : a universal constant known as the constant of gravitation,
 $6.673(10^{-11}) \text{ m}^3/(\text{kg}.\text{s}^2)$.

M : the masses of the earth

m : the masses of the body on the surface of earth

r : distance between the earth's centre and the body

The standard value **for** gravitational acceleration 'g' is

$$\text{SI units} \quad g = 9.806 \, 65 \text{ m/s}^2$$

$$\text{U.S. units} \quad g = 32.1740 \text{ ft/sec}^2$$

This value is at sea level and at a 45° latitude.

Units

- In mechanics we use **four fundamental quantities** called dimensions.
- These are **length (space), mass, force, and time**.

QUANTITY	DIMENSIONAL SYMBOL
Mass	M
Length	L
Time	T
Force	F

- The **units** used to measure these quantities cannot all be chosen independently
- It must be consistent with **Newton's second law**

SI Units

- The **International System of Units**, abbreviated SI (from the French, *Système International d'Unités*),
- Accepted throughout the world, and is a **modern version of the metric system**.
- By international agreement, SI units **will in time replace** other systems.

Base units

QUANTITY	DIMENSIONAL SYMBOL
Mass	M
Length	L
Time	T
Force	F

SI UNITS		
UNIT		SYMBOL
Base units	kilogram	kg
	meter	m
	second	s
	newton	N

Derived Unit

- **Newton** (N) is derived unit of force
- $\text{Force (N)} = \text{mass (kg)} \times \text{acceleration (m/s}^2\text{)}$
 - $N = \text{kg} \cdot \text{m/s}^2$ (MLT⁻²)
 - Thus, **1 newton is the force required** to give a mass of 1 kg an acceleration of 1 m/s^2 .

QUANTITY	DIMENSIONAL SYMBOL
Mass	M
Length	L
Time	T
Force	F

SI UNITS		
	UNIT	SYMBOL
Base units	kilogram	kg
	meter	m
	second	s
	newton	N

U.S. Customary Units

- The U.S. customary, or British system of units, also called **the Foot-Pound-Second (FPS) system**,
- It has common system in both business and industry in most of the **English-speaking countries**.
- System will in time be **replaced by SI units**,

QUANTITY	DIMENSIONAL SYMBOL
Mass	M
Length	L
Time	T
Force	F

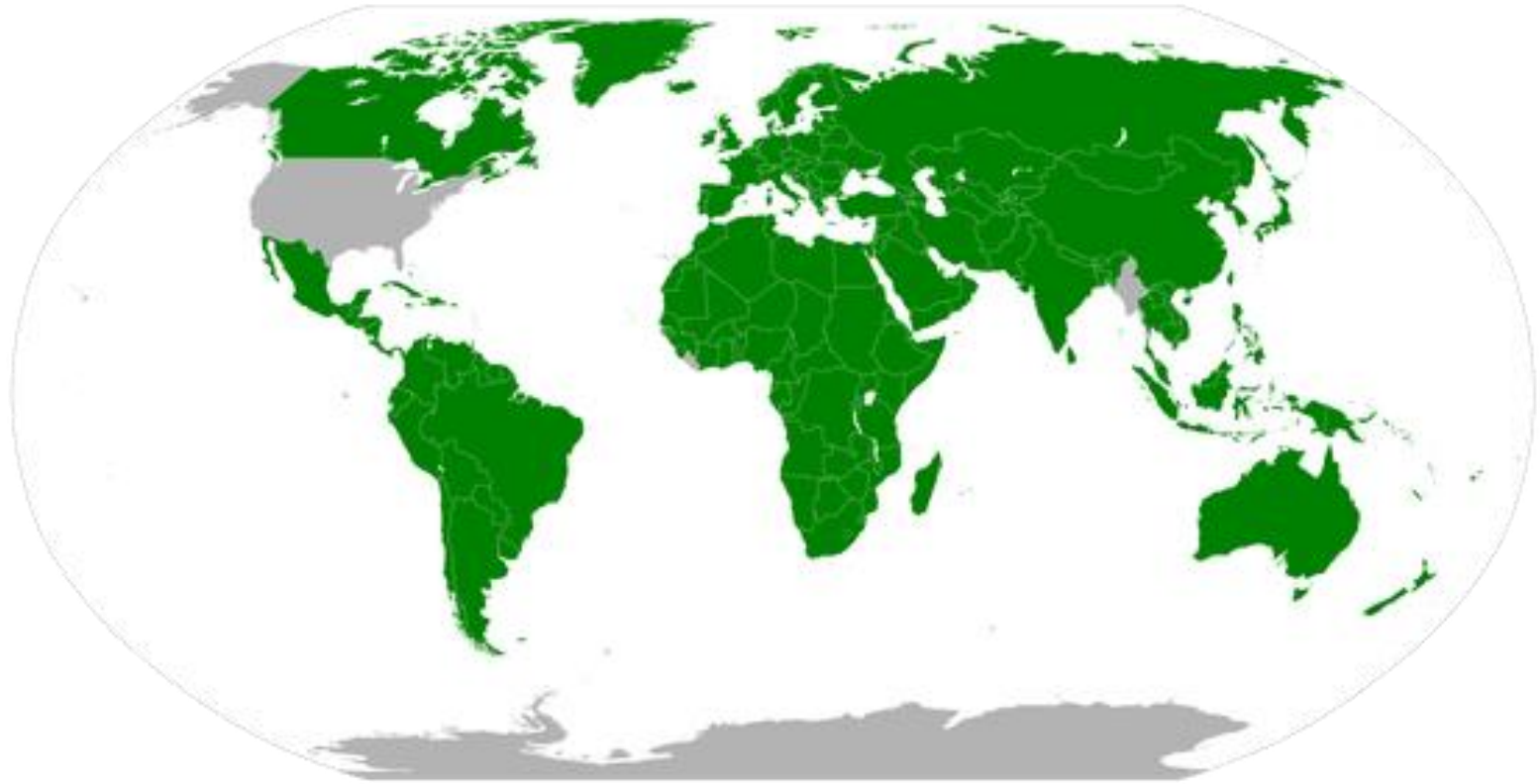
U.S. CUSTOMARY UNITS	
UNIT	SYMBOL
slug	—
Base units	foot
	ft
	sec
	second
	lb
	pound

U.S. Customary Units

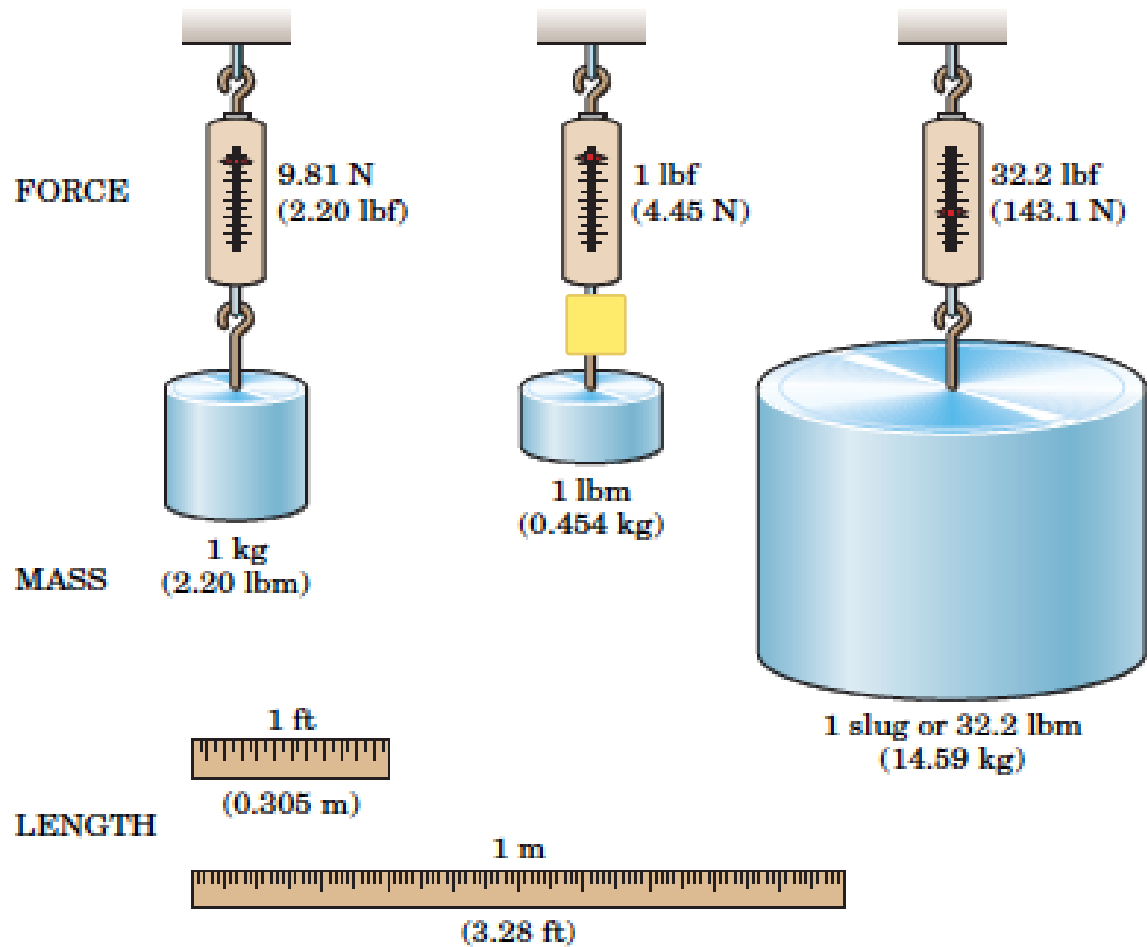
- *pound* is unit of force (basic unit)
- *slug* is unit of mass (derived unit)
- $\text{Force (lb)} = \text{mass (slug)} \times \text{acceleration (ft/sec}^2\text{)}$
 - $\text{slug} = \text{lb} \cdot \text{sec}^2 / \text{ft (FL}^{-1}\text{S}^2\text{)}$
- Therefore, 1 *slug* is the mass which is given an acceleration of 1 ft/s^2 when acted on by a force of 1 *lb*.

QUANTITY	DIMENSIONAL SYMBOL
Mass	M
Length	L
Time	T
Force	F

U.S. CUSTOMARY UNITS	
UNIT	SYMBOL
slug	—
Base units	foot
	second
	pound
	ft
	sec
	lb



Unit Conversion



Primary Standards

- **Primary standards** for the measurements of **mass**, **length**, and **time** have been established by international agreement using some reference



Mass

- The kilogram is defined as the mass of a **specific platinum iridium cylinder**
- It is kept at the **International Bureau of Weights and Measures** near Paris, France.
- An accurate copy is kept in the United States at the **National Institute of Standards and Technology** (NIST).



The standard kilogram

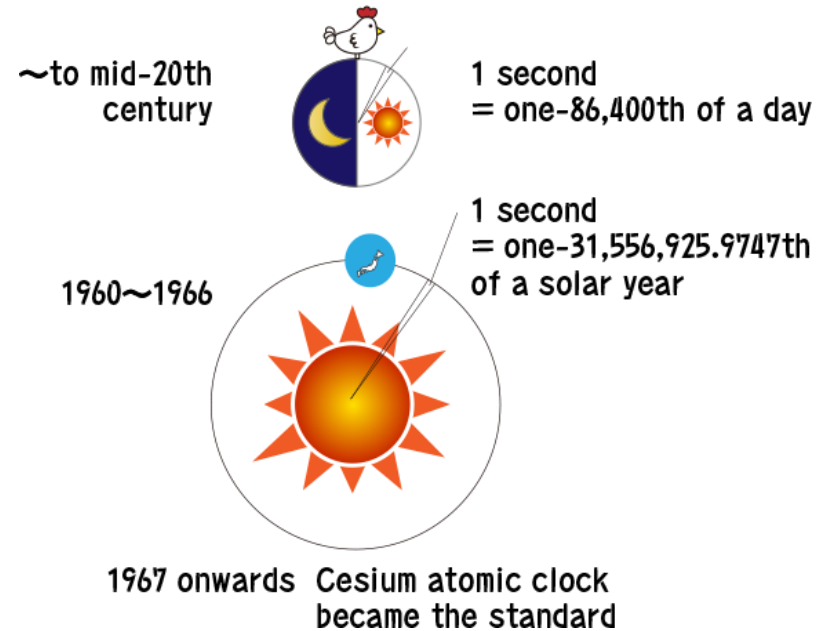
Length

- Meter: Originally, **one ten-millionth** of the **distance from the pole to the equator** along the meridian through Paris
- Later defined as length of a **specific platinum-iridium bar** kept at the **International Bureau of Weights and Measures**.
- The **difficulty of accessing** the bar and **reproducing** accurate measurements
- length for the meter is now defined as **distance traveled by light in a vacuum in $(1/299\,792\,458)$ second**

Time

- The second was originally defined as the fraction $1/(86\,400)$ of the mean solar day.
- Irregularities in the earth's rotation led to difficulties, and a more accurate and reproducible standard has been adopted.
- Atomic clock: The duration of 9192 631 770 periods of the radiation of a specific state of the Cesium-133 atom.

The changing definition of a second



Problem Solving in Statics

Part A: Formulate the problem

Part B: Develop the solution

Part A: Formulate the problem

- a) State the **given data**.
- b) State the **desired result**.
- c) State your **assumptions** and **approximations**.

Part B: Develop the solution

- (a) **Draw diagrams** that you need to understand the relationships.
- (b) **State the governing principles** to be applied for solution.
- (c) **Make calculations.**
- (d) Check **consistency of calculations** with the accuracy of data.
- (e) **Use consistent units** throughout calculations.
- (f) Ensure that answers are reasonable in terms of **magnitudes, directions, common sense**, etc.
- (g) Draw **conclusions.**

Making Appropriate Assumptions

- While constructing an **idealized mathematical model** for a given engineering problem, certain **approximations** will always be involved.
- It is often necessary to **neglect small distances, angles, or forces** compared with large distances, angles, or forces for simplification.
- **Force** is distributed over a **small area** may be considered it to be a **concentrated force**

Using Graphics

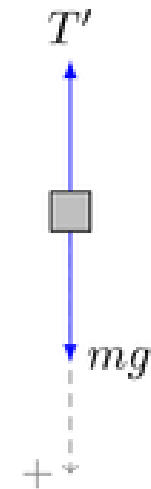
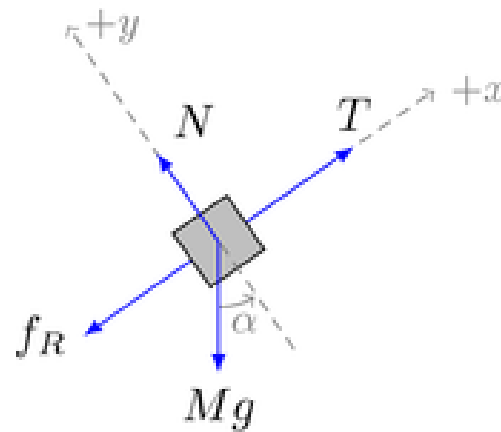
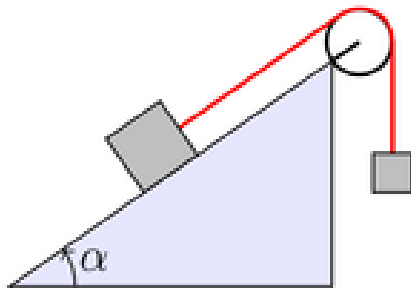
Graphics is an important analytical tool for three reasons:

1. Graphical representation helps in physical interpretation, especially when we must visualize three-dimensional problems.
2. Graphical solution to problems is more easy and practical than mathematical solution and an aid in our thought processes.
3. Charts or graphs are valuable aids for representing results in a form which is easy to understand.

The Free-Body Diagram

- Used in applying the principles of mechanics to **analyse forces acting on a body**.
- it is essential to **isolate the body** from all other bodies to get **complete and accurate account** of all **forces** acting on this
- The diagram of such an isolated body with all external forces acting on it is called a **freebody diagram**.

Free Body Diagram



Numerical Values versus Symbols

- We may use **numerical values to represent quantities**, or use **algebraic symbols and leave the answer as a formula**.
- Numerical Values:
 - Useful when we need to **know the magnitude** of each term.
- Symbols
 - helps to focus on the **connection** between the **physical situation** and its related **mathematical description**.
 - Symbolic solution be used for obtaining **answers to the same type of problem**, but having different units or numerical values.
 - A symbolic solution **enables** us to make a **dimensional check at every step**, which is more difficult when numerical values used

Dimensional Homogeneity

- In any equation representing a physical situation, the **dimensions of every term** on both sides of the equation **must be the same**.
- This property is called dimensional homogeneity

Solution Methods

- Obtain **mathematical solutions by hand**.
- Obtain **graphical solutions** for certain problems.
- Solve problems **by computer** when
 - a **large number of equations** must be solved, a parameter variation must be studied,
 - an **intractable equation** must be solved.

THANK YOU